

Organizing Data Using Frequency Distribution Tables

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A frequency distribution table is used to organize raw data into a meaningful form.

Frequency

Frequency tells how many times a value appears in a data set.

5, 6, 5, 7, 5

The frequency of 5 is 3.

Tally Mark

A **tally mark** is used to keep track of the number of occurrences of each value.

Value	Tally
5	
6	
7	

Categories and Scores

The values being counted are called **categories**, **scores**, or **data values**.

For example, in the data

5, 6, 5, 7, 6

the data values are 5, 6, and 7.

Organizing Data

Organizing data into a frequency table makes it easier to compare values and analyze patterns.

Value	Frequency
5	2
6	2
7	1

Total Frequency

The sum of all frequencies is called the **total frequency**.

It is equal to the number of data items.

$$2 + 2 + 1 = 5$$

Therefore, the total frequency is 5.

Uses of Frequency Tables

Frequency distribution tables are commonly used before constructing graphs such as bar graphs and pie graphs.

Raw Data $\longrightarrow$ Frequency Table $\longrightarrow$ Graph

Examples

Example 1

Organize the data into a frequency distribution table.

5, 6, 5, 7, 8, 6, 5, 7

Value	Frequency
5	3
6	2
7	2
8	1

Example 2

Create a frequency distribution table.

1, 2, 1, 3, 2, 1, 4, 2

Value	Frequency
1	3
2	3
3	1
4	1

Example 3

Make a frequency distribution table.

10, 11, 10, 12, 13, 11, 10, 12

Value	Frequency
10	3
11	2
12	2
13	1

Practice Exercises

Practice 1

Organize the following data into a frequency distribution table.

5, 6, 5, 7, 8, 6, 5, 7, 6, 8, 5, 7, 6, 5, 8

Practice 2

Create a frequency distribution table for these daily study hours.

1, 2, 1, 3, 2, 1, 4, 2, 3, 1, 2, 4, 3, 2, 1

Practice 3

Make a frequency distribution table for the following numbers.

10, 11, 10, 12, 13, 11, 10, 12, 11, 13, 10, 11

Practice 4

Construct a frequency distribution table.

15, 16, 15, 17, 18, 16, 15, 17, 16, 18, 15, 17

Practice 5

Create a frequency distribution table.

4, 5, 6, 4, 5, 4, 7, 6, 5, 4, 6, 7, 5, 4, 6

Activities

Activity 1

Construct a frequency distribution table for the following favorite numbers.

1, 2, 3, 1, 4, 3, 2, 1, 4, 3, 1, 2, 3, 4, 1

Activity 2

Organize the following shoe sizes into a frequency distribution table.

37, 38, 39, 38, 37, 40, 39, 38, 37, 39, 40, 38

Activity 3

Organize the following ages into a frequency distribution table.

11, 12, 11, 13, 12, 11, 14, 13, 12, 11, 14, 13

Activity 4

Organize the following data.

24, 25, 26, 24, 27, 26, 25, 24, 27, 26, 24, 25

Activity 5

Organize the following test scores into a frequency distribution table.

65, 70, 75, 70, 65, 80, 75, 70, 65, 75, 80, 70

Thank you!